Multivendor Network Engineering Lab Portfolio

Student: Emmanuel Ifada
Focus: Multivendor Routing & Multicast Networking
Platforms: Cisco, Juniper, Nokia, Linux Routers

Technical Skills & Competencies

- Multi-vendor routing (Cisco IOS, Juniper Junos, Nokia SR OS/SR Linux)
- Dynamic routing protocols: OSPF, IS-IS, BGP, RIP
- Layer 2 technologies: VLANs, STP
- Multicast routing (PIM, RP configuration, video streaming validation)
- Network troubleshooting and validation (ping, routing tables, protocol debugging)
- Network emulation using containerlab
- Linux-based network testing and traffic validation

Executive Summary – Networking Lab Portfolio

This portfolio documents a series of hands-on networking laboratory exercises conducted to develop practical, real-world skills in enterprise and service-provider networking environments. The work focuses on configuring, validating, and troubleshooting networks using a multivendor approach, including Cisco, Juniper, Nokia, and Linux-based routing platforms.

The labs cover core networking technologies such as VLANs and inter-VLAN routing, Spanning Tree Protocol (STP), static and dynamic routing protocols (RIP, OSPF, IS-IS, and BGP), Network Address Translation (NAT), and multicast routing. Each lab was designed to reinforce theoretical concepts through implementation, verification, and troubleshooting, closely simulating real production network scenarios.

A strong emphasis was placed on validation and testing. Connectivity was verified using host-to-host pings, loopback reachability, routing table inspection, and multicast video streaming, ensuring that configurations were functional rather than purely theoretical. Configuration errors and design issues encountered during the labs were identified and corrected, reinforcing a structured troubleshooting methodology.

This portfolio demonstrates proficiency in vendor-neutral networking concepts, adaptability across different router operating systems, and the ability to reason about network behavior rather than rely on memorized commands. It is intended to support applications for networking internships, co-op placements, and entry-level roles such as NOC technician or junior network engineer, where hands-on skills, troubleshooting ability, and a strong foundation in routing and switching are essential.

Reason and Purpose of the Laboratory Work

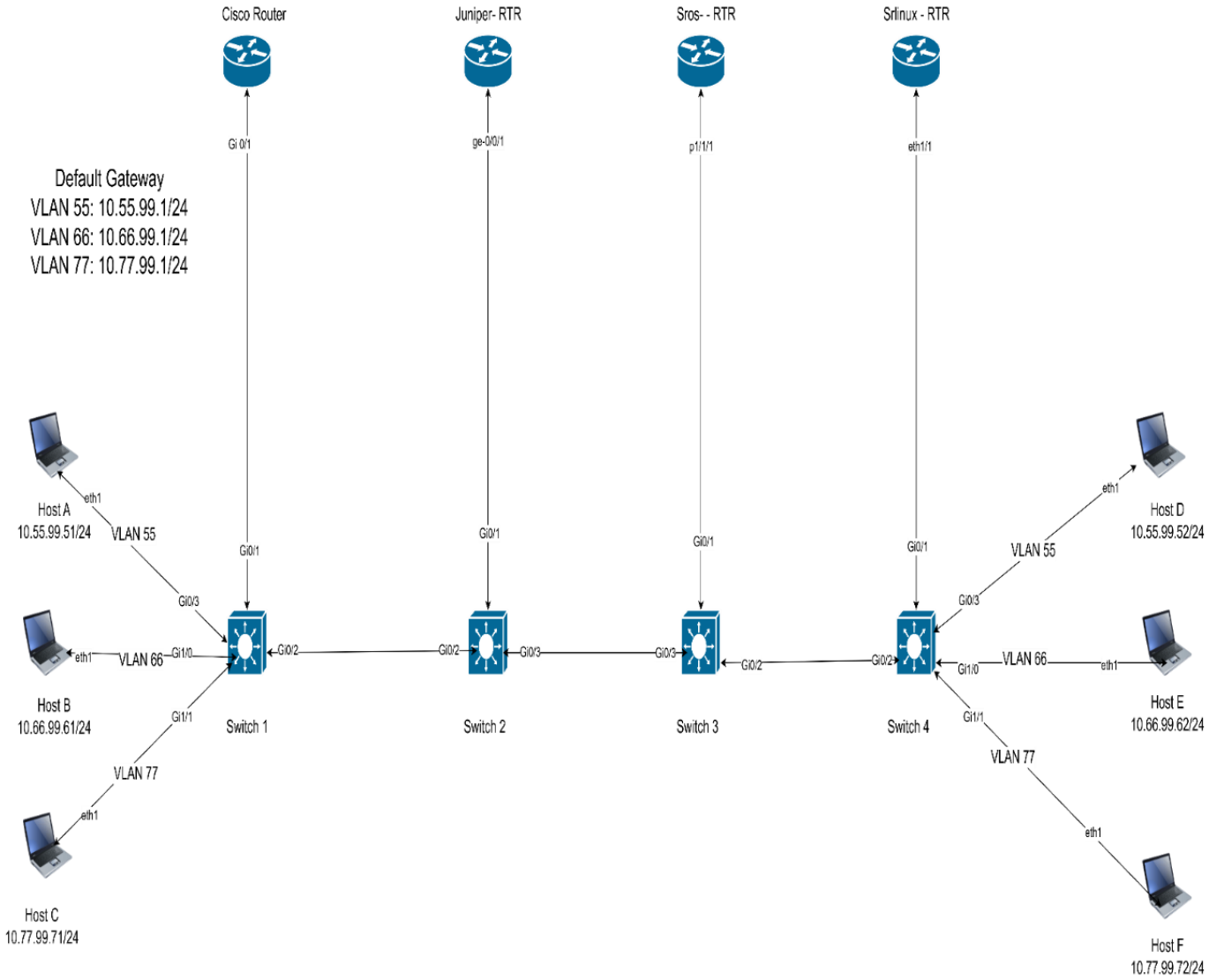
The laboratory work was conducted to translate theoretical networking concepts into practical, real-world implementations, with a strong emphasis on developing vendor-neutral skills. The primary objective was to gain hands-on experience configuring, validating, and troubleshooting enterprise-level routing and multicast networking across multiple routing platforms.

To reflect real production environments, the labs were intentionally designed using a multivendor approach, incorporating Cisco, Juniper, Nokia, and Linux-based routers. This approach helped bridge the gap between theory and practice while demonstrating how core networking principles remain consistent despite differences in vendor-specific operating systems and configuration styles.

Through this work, I developed a deeper understanding of network design, interoperability, and the importance of adaptability when working in heterogeneous network environments.

1. VLAN & Inter-VLAN Routing

- Configured VLAN segmentation and inter-VLAN routing across a multi-vendor environment
- Implemented trunking and VLAN tagging to enable communication between network segments
- Validated Layer 2 and Layer 3 connectivity using end-to-end ping tests
- Troubleshot connectivity issues caused by incorrect VLAN assignments and trunk configurations
- Ensured proper default gateway configuration for inter-VLAN communication



Host A to Host B

```
int@Host-A:~$ ping 10.66.99.61
PING 10.66.99.61 (10.66.99.61): 56 data bytes
64 bytes from 10.66.99.61: icmp_seq=0 ttl=63 time=10.148 ms
64 bytes from 10.66.99.61: icmp_seq=1 ttl=63 time=9.506 ms
64 bytes from 10.66.99.61: icmp_seq=2 ttl=63 time=9.221 ms
64 bytes from 10.66.99.61: icmp_seq=3 ttl=63 time=9.526 ms
64 bytes from 10.66.99.61: icmp_seq=4 ttl=63 time=7.292 ms
^C--- 10.66.99.61 ping statistics ---
5 packets transmitted, 5 packets received, 0% packet loss
round-trip min/avg/max/stddev = 7.292/9.139/10.148/0.972 ms
```

Host B to Host C

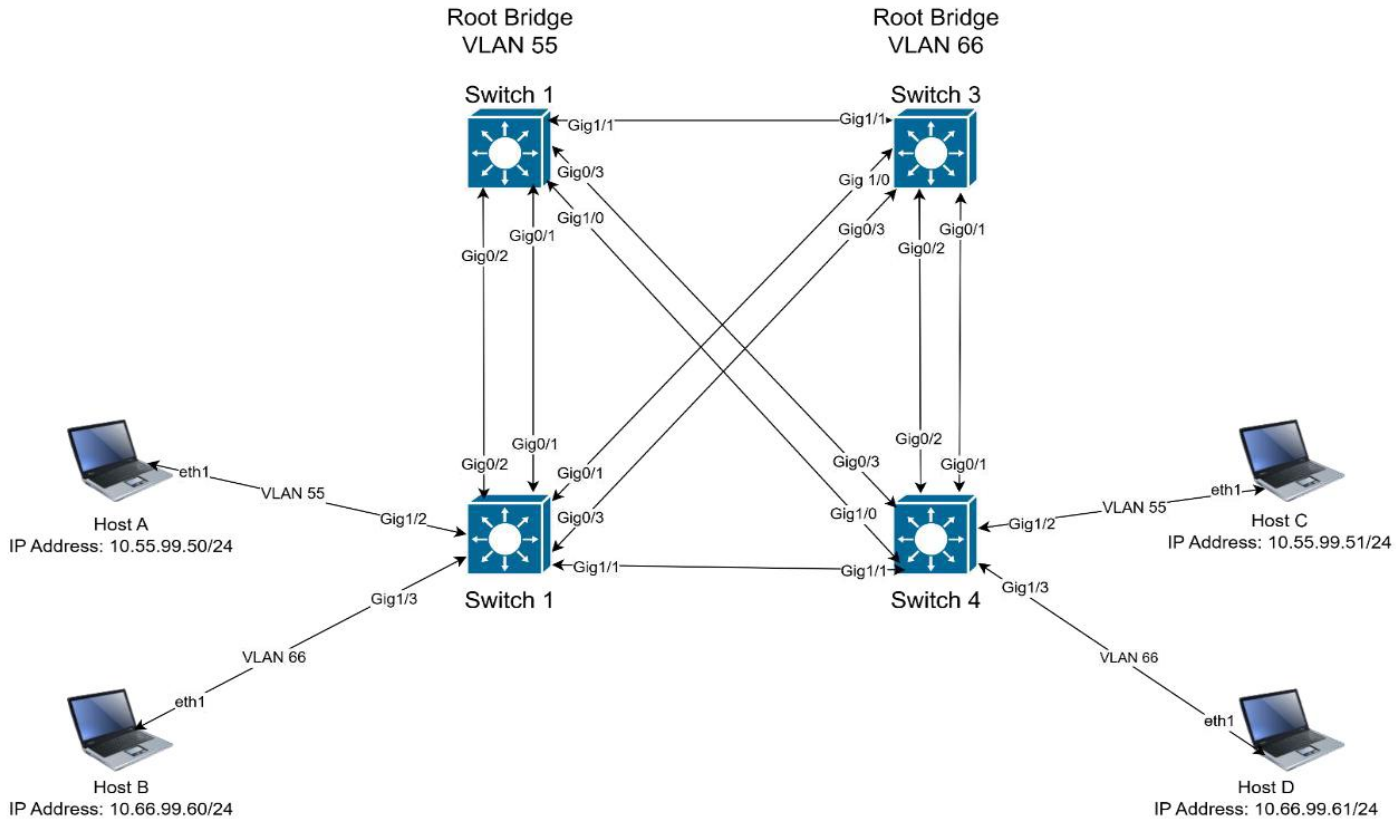
```
mint@Host-B:~$ ping 10.77.99.71
PING 10.77.99.71 (10.77.99.71): 56 data bytes
64 bytes from 10.77.99.71: icmp_seq=0 ttl=63 time=6.867 ms
64 bytes from 10.77.99.71: icmp_seq=1 ttl=63 time=6.993 ms
64 bytes from 10.77.99.71: icmp_seq=2 ttl=63 time=8.443 ms
64 bytes from 10.77.99.71: icmp_seq=3 ttl=63 time=7.844 ms
^C--- 10.77.99.71 ping statistics ---
4 packets transmitted, 4 packets received, 0% packet loss
round-trip min/avg/max/stddev = 6.867/7.537/8.443/0.644 ms
```

Host C to Host D

```
mint@Host-C:~$ ping 10.55.99.52
PING 10.55.99.52 (10.55.99.52): 56 data bytes
64 bytes from 10.55.99.52: icmp_seq=0 ttl=63 time=16.584 ms
64 bytes from 10.55.99.52: icmp_seq=1 ttl=63 time=15.904 ms
64 bytes from 10.55.99.52: icmp_seq=2 ttl=63 time=19.356 ms
64 bytes from 10.55.99.52: icmp_seq=3 ttl=63 time=16.002 ms
64 bytes from 10.55.99.52: icmp_seq=4 ttl=63 time=18.030 ms
^C--- 10.55.99.52 ping statistics ---
5 packets transmitted, 5 packets received, 0% packet loss
round-trip min/avg/max/stddev = 15.904/17.175/19.356/1.329 ms
```

2. Spanning Tree Protocol (STP)

- Configured STP to prevent Layer 2 loops while maintaining network redundancy
- Analyzed root bridge election and optimized traffic paths by adjusting bridge priorities
- Verified port roles and states to ensure correct topology behavior
- Troubleshot suboptimal traffic flow caused by unintended root bridge selection
- Validated network stability after topology changes



Host C to Host G

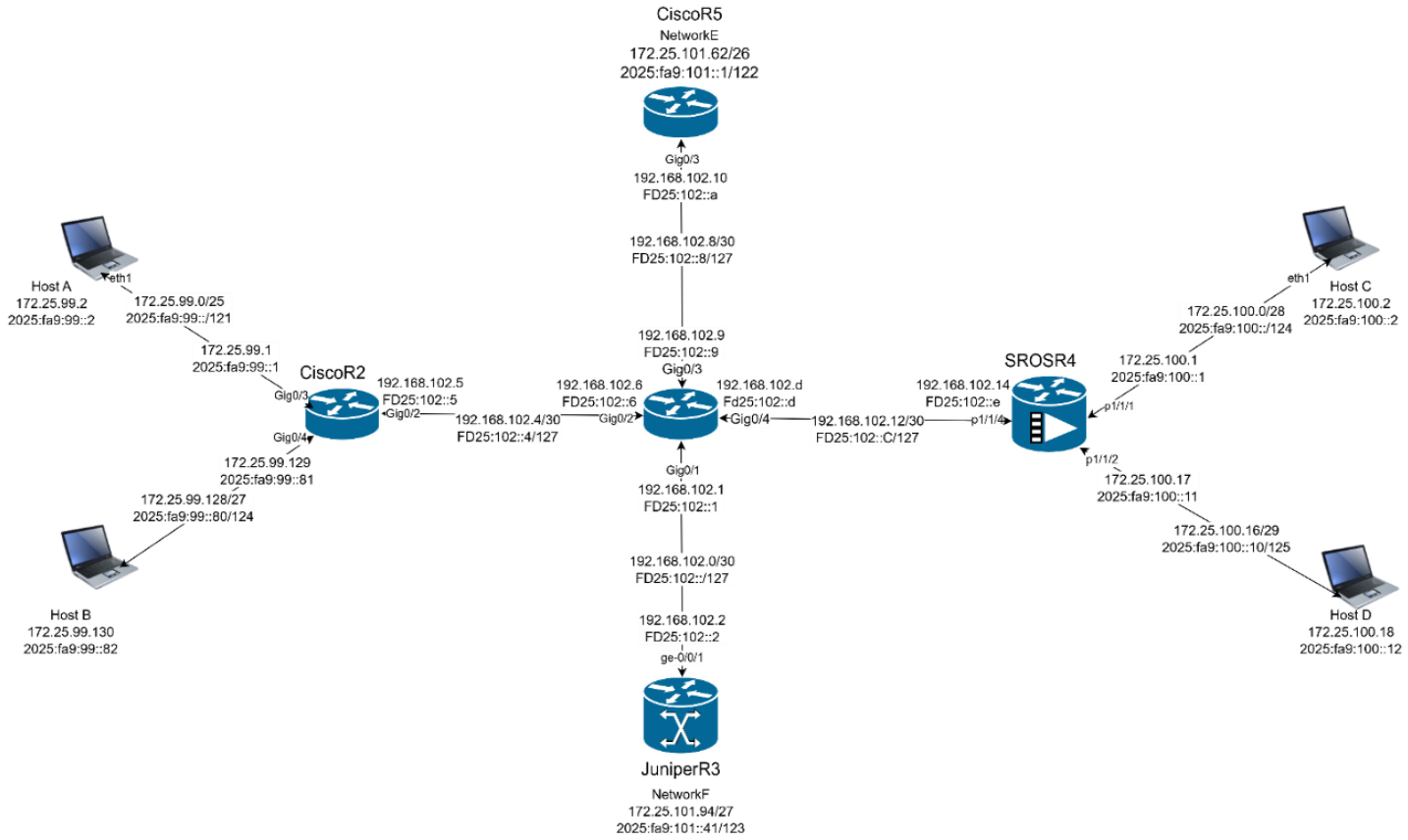
```
mint@Host-C:~$ ping 10.77.99.71
PING 10.77.99.71 (10.77.99.71): 56 data bytes
64 bytes from 10.77.99.71: icmp_seq=0 ttl=64 time=24.416 ms
64 bytes from 10.77.99.71: icmp_seq=1 ttl=64 time=8.716 ms
64 bytes from 10.77.99.71: icmp_seq=2 ttl=64 time=5.607 ms
64 bytes from 10.77.99.71: icmp_seq=3 ttl=64 time=9.493 ms
64 bytes from 10.77.99.71: icmp_seq=4 ttl=64 time=9.871 ms
64 bytes from 10.77.99.71: icmp_seq=5 ttl=64 time=8.113 ms
64 bytes from 10.77.99.71: icmp_seq=6 ttl=64 time=8.314 ms
^C--- 10.77.99.71 ping statistics ---
7 packets transmitted, 7 packets received, 0% packet loss
round-trip min/avg/max/stddev = 5.607/10.647/24.416/5.763 ms
```

Host D to Host G

```
mint@Host-D:~$ ping 10.88.99.81
PING 10.88.99.81 (10.88.99.81): 56 data bytes
64 bytes from 10.88.99.81: icmp_seq=0 ttl=64 time=22.508 ms
64 bytes from 10.88.99.81: icmp_seq=1 ttl=64 time=10.741 ms
64 bytes from 10.88.99.81: icmp_seq=2 ttl=64 time=9.287 ms
64 bytes from 10.88.99.81: icmp_seq=3 ttl=64 time=6.873 ms
64 bytes from 10.88.99.81: icmp_seq=4 ttl=64 time=8.364 ms
64 bytes from 10.88.99.81: icmp_seq=5 ttl=64 time=9.669 ms
64 bytes from 10.88.99.81: icmp_seq=6 ttl=64 time=10.164 ms
^C--- 10.88.99.81 ping statistics ---
7 packets transmitted, 7 packets received, 0% packet loss
round-trip min/avg/max/stddev = 6.873/11.087/22.508/4.808 ms
```

3. Routing Information Protocol (RIP)

- Configured RIP for dynamic routing and route advertisement between routers
- Analyzed routing behavior including hop-count limitations and convergence time
- Troubleshot route propagation issues caused by incorrect network statements
- Ensured version compatibility and proper routing updates between devices
- Evaluated RIP limitations in scalability and performance



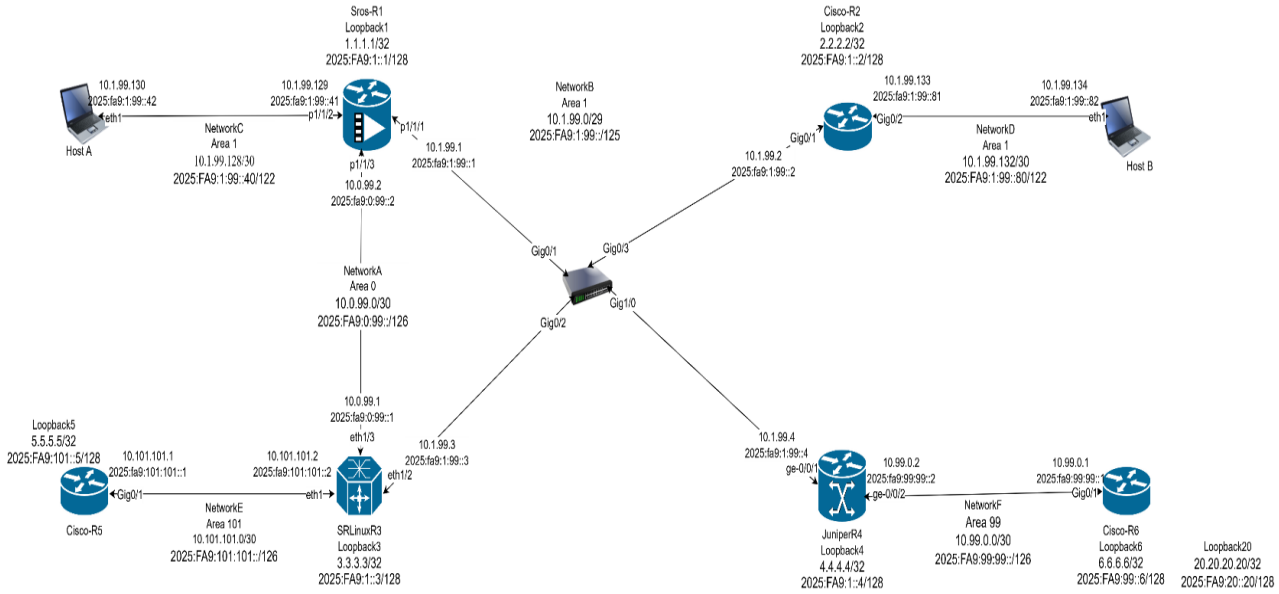
Host A to Host B

```
mint@Host-A:~$ ping 172.25.99.130
PING 172.25.99.130 (172.25.99.130): 56 data bytes
64 bytes from 172.25.99.130: icmp_seq=0 ttl=63 time=1.298 ms
64 bytes from 172.25.99.130: icmp_seq=1 ttl=63 time=1.717 ms
64 bytes from 172.25.99.130: icmp_seq=2 ttl=63 time=0.936 ms
64 bytes from 172.25.99.130: icmp_seq=3 ttl=63 time=1.810 ms
^C--- 172.25.99.130 ping statistics ---
4 packets transmitted, 4 packets received, 0% packet loss
round-trip min/avg/max/stddev = 0.936/1.440/1.810/0.349 ms
```

```
mint@Host-A:~$ ping6 -n 2025:fa9:99::82
PING 2025:fa9:99::82 (2025:fa9:99::82): 56 data bytes
64 bytes from 2025:fa9:99::82: icmp_seq=0 ttl=63 time=33.416 ms
64 bytes from 2025:fa9:99::82: icmp_seq=1 ttl=63 time=1.360 ms
64 bytes from 2025:fa9:99::82: icmp_seq=2 ttl=63 time=1.074 ms
64 bytes from 2025:fa9:99::82: icmp_seq=3 ttl=63 time=3.313 ms
64 bytes from 2025:fa9:99::82: icmp_seq=4 ttl=63 time=1.302 ms
^C--- 2025:fa9:99::82 ping statistics ---
5 packets transmitted, 5 packets received, 0% packet loss
round-trip min/avg/max/stddev = 1.074/8.093/33.416/12.687 ms
```

4. Open Shortest Path First (OSPF)

- Configured OSPF with multiple areas to support hierarchical network design
- Established and verified OSPF neighbor adjacencies across devices
- Troubleshot adjacency failures caused by mismatched hello and dead intervals
- Validated routing table entries and LSDB synchronization
- Analyzed OSPF behavior in enterprise network environments



Loopback 20 to Host B

```
Ifada-Cisco-R6#ping 10.1.99.134
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.1.99.134, timeout is 2 seconds:
!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 9/10/12 ms
```

```
Ifada-Cisco-R6#ping 2025:fa9:1:99::82
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 2025:FA9:1:99::82, timeout is 2 seconds:
!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 6/7/10 ms
```

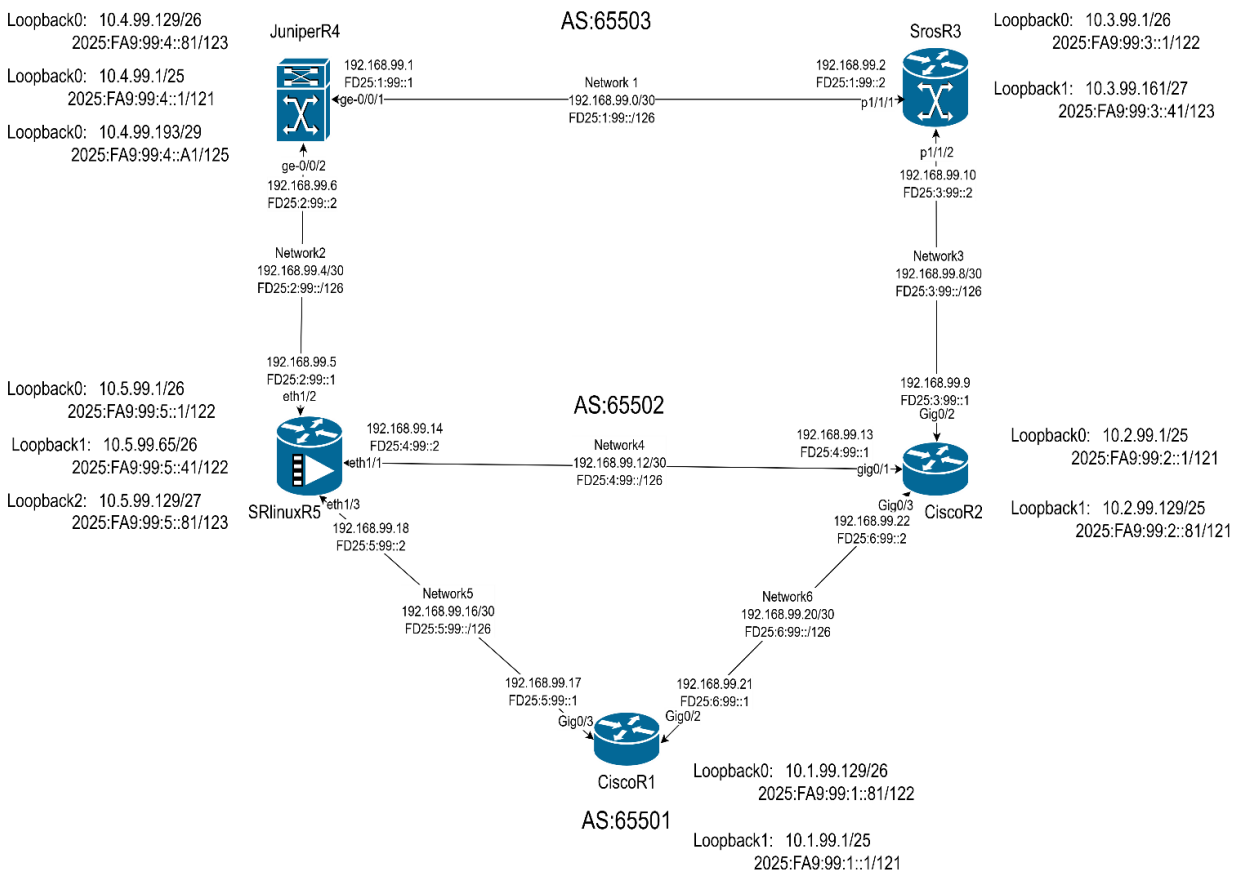
Host B to Loopback 20

```
mint@Host-B:~$ traceroute 20.20.20.20
traceroute to 20.20.20.20 (20.20.20.20), 30 hops max, 60 byte packets
 1  10.1.99.133 (10.1.99.133)  3.572 ms  8.743 ms  13.276 ms
 2  10.1.99.4 (10.1.99.4)  10.958 ms  15.921 ms  17.474 ms
 3  10.99.0.1 (10.99.0.1)  22.576 ms * *
```

```
mint@Host-B:~$ traceroute 2025:fa9:20::20
traceroute to 2025:fa9:20::20 (2025:fa9:20::20), 30 hops max, 80 byte packets
 1  2025:fa9:1:99::81 (2025:fa9:1:99::81)  3.064 ms  7.636 ms  9.638 ms
 2  2025:fa9:1:99::4 (2025:fa9:1:99::4)  17.229 ms  18.395 ms  23.049 ms
 3  2025:fa9:99:99::1 (2025:fa9:99:99::1)  24.963 ms  27.584 ms  36.422 ms
```

5. Border Gateway Protocol (BGP)

- Configured BGP sessions between autonomous systems for inter-domain routing
- Verified route exchange and analyzed BGP path selection behavior
- Troubleshot session establishment issues caused by incorrect AS configurations
- Ensured proper neighbor relationships and TCP connectivity
- Evaluated BGP's role in policy-based routing and large-scale networks



Loopback B to Loopback H

```
Ifada-Cisco-R1#ping 10.4.99.1
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.4.99.1, timeout is 2 seconds:
!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 5/6/7 ms
```

```
Ifada-Cisco-R1#ping 2025:fa9:99:4::1
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 2025:FA9:99:4::1, timeout is 2 seconds:
 !!!!!
 Success rate is 100 percent (5/5), round-trip min/avg/max = 2/2/4 ms
```

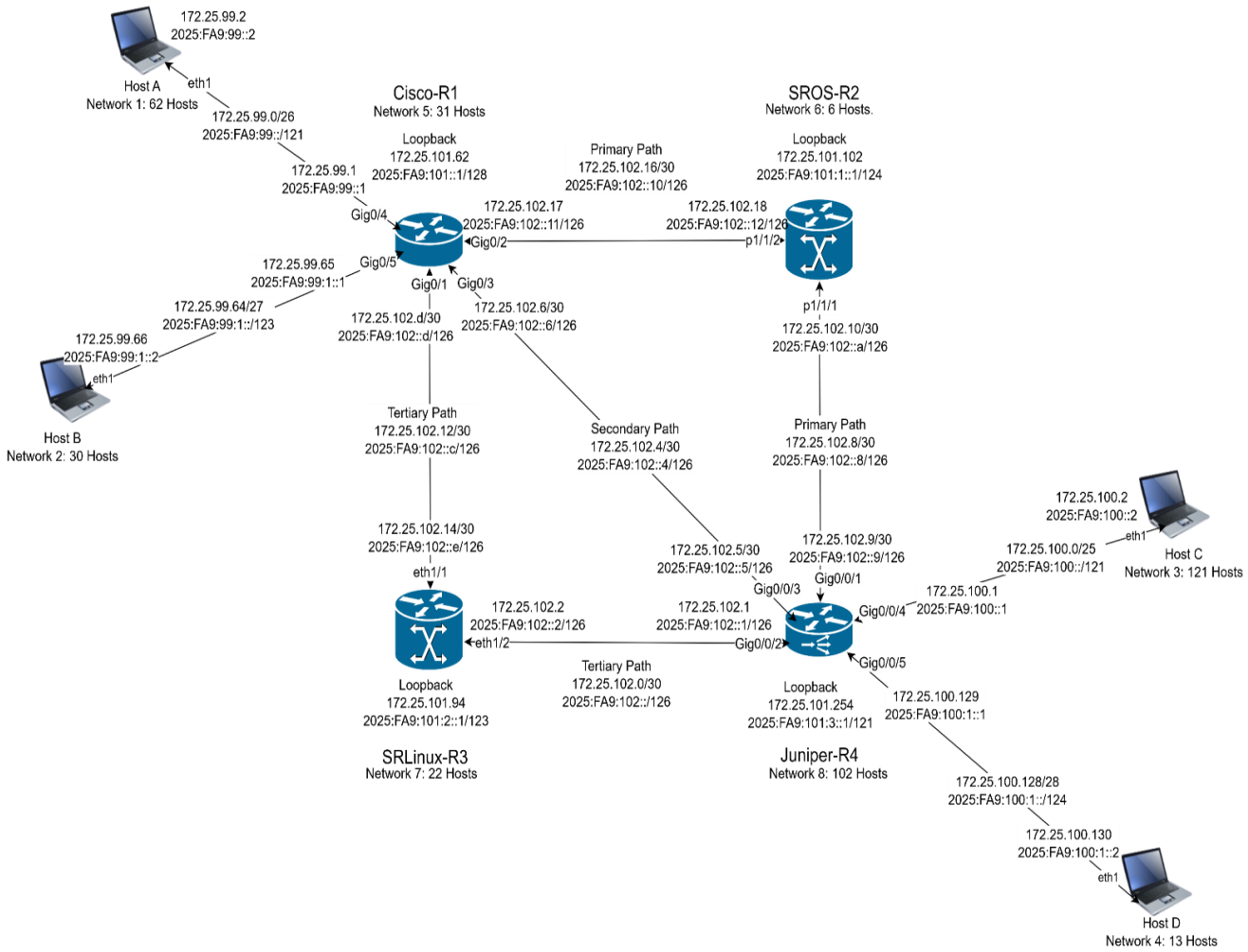
R3 to Loopback B

```
*A:Ifada-SROS# traceroute 10.1.99.1
traceroute to 10.1.99.1, 30 hops max, 40 byte packets
  1  192.168.99.9 (192.168.99.9)    3.76 ms  4.03 ms  3.95 ms
  2  192.168.99.21 (192.168.99.21)    4.78 ms
```

```
*A:Ifada-SROS# traceroute 2025:fa9:99:1::1
traceroute to 2025:fa9:99:1::1, 30 hops max, 60 byte packets
  1  fd25:3:99::1 (fd25:3:99::1)    4.54 ms  3.90 ms  4.24 ms
  2  fd25:6:99::1 (fd25:6:99::1)    5.21 ms  5.38 ms  5.97 ms
```

6. Static Routing

- Configured static routes to control traffic flow across the network
- Verified route reachability and ensured correct next-hop configuration
- Troubleshot routing failures caused by unreachable next-hop addresses
- Demonstrated use of static routing for predictable and controlled environments
- Evaluated static routing as a backup to dynamic routing protocols



Host B to Host D

```
mint@Host-B:~$ ping 172.25.100.130
PING 172.25.100.130 (172.25.100.130): 56 data bytes
64 bytes from 172.25.100.130: icmp_seq=0 ttl=61 time=5.025 ms
64 bytes from 172.25.100.130: icmp_seq=1 ttl=61 time=6.990 ms
64 bytes from 172.25.100.130: icmp_seq=2 ttl=61 time=6.001 ms
64 bytes from 172.25.100.130: icmp_seq=3 ttl=61 time=4.962 ms
64 bytes from 172.25.100.130: icmp_seq=4 ttl=61 time=4.917 ms
^C--- 172.25.100.130 ping statistics ---
5 packets transmitted, 5 packets received, 0% packet loss
round-trip min/avg/max/stddev = 4.917/5.579/6.990/0.812 ms
```

```
mint@Host-B:~$ ping6 -n 2025:fa9:100:1::2
PING 2025:fa9:100:1::2 (2025:fa9:100:1::2): 56 data bytes
64 bytes from 2025:fa9:100:1::2: icmp_seq=0 ttl=61 time=3.751 ms
64 bytes from 2025:fa9:100:1::2: icmp_seq=1 ttl=61 time=3.232 ms
64 bytes from 2025:fa9:100:1::2: icmp_seq=2 ttl=61 time=5.870 ms
64 bytes from 2025:fa9:100:1::2: icmp_seq=3 ttl=61 time=5.684 ms
64 bytes from 2025:fa9:100:1::2: icmp_seq=4 ttl=61 time=5.849 ms
64 bytes from 2025:fa9:100:1::2: icmp_seq=5 ttl=61 time=6.640 ms
^C--- 2025:fa9:100:1::2 ping statistics ---
6 packets transmitted, 6 packets received, 0% packet loss
round-trip min/avg/max/stddev = 3.232/5.171/6.640/1.235 ms
```

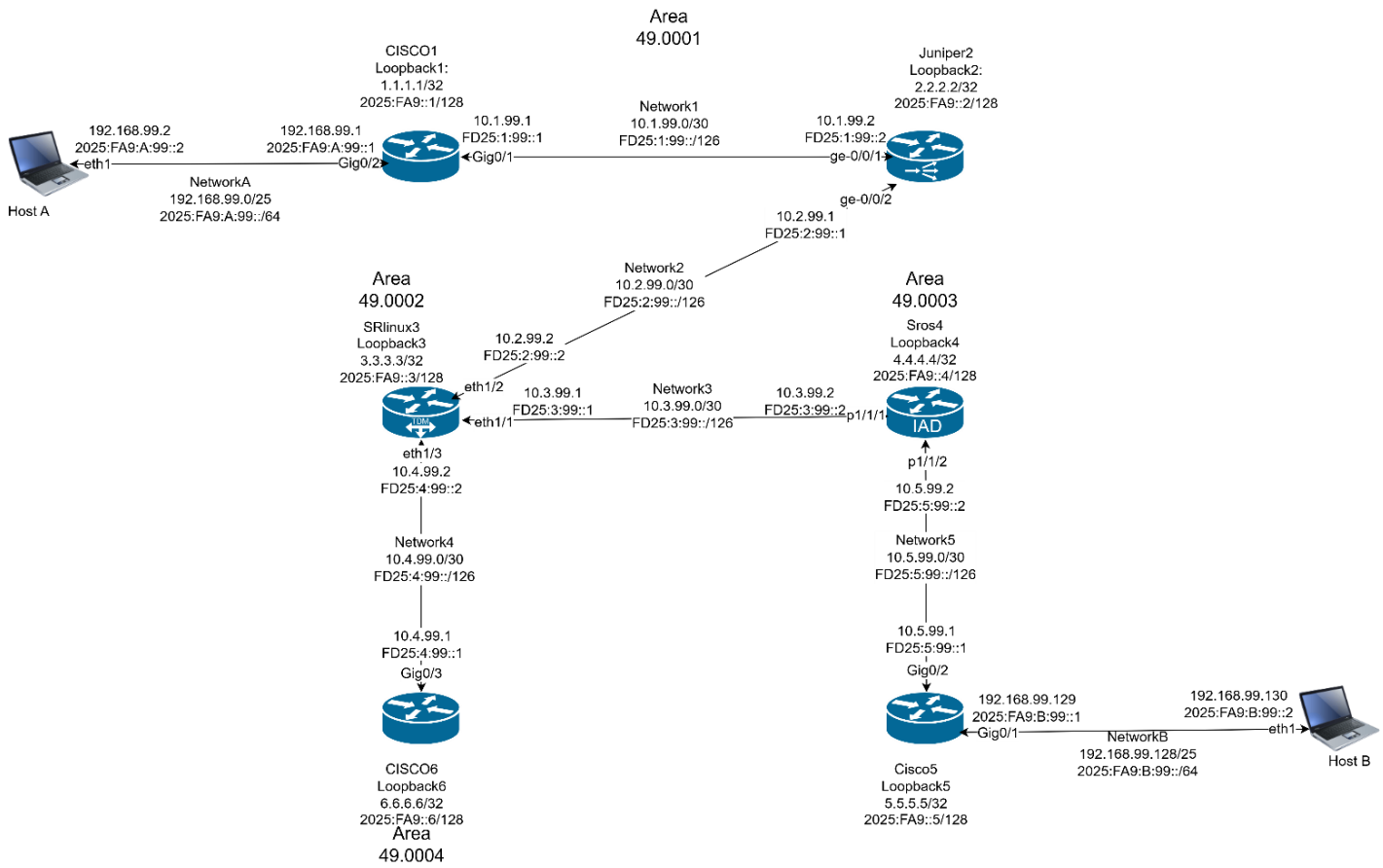
Host D to Host B

```
mint@Host-D:~$ traceroute 172.25.99.66
traceroute to 172.25.99.66 (172.25.99.66), 30 hops max, 60 byte packets
 1  172.25.100.129 (172.25.100.129)  1.294 ms  1.245 ms  1.232 ms
 2  172.25.102.10 (172.25.102.10)  3.012 ms  3.004 ms  2.988 ms
 3  172.25.102.17 (172.25.102.17)  15.448 ms  16.925 ms  20.133 ms
 4  172.25.99.66 (172.25.99.66)  15.405 ms  25.037 ms  25.033 ms
```

```
mint@Host-D:~$ traceroute 2025:fa9:99:1::2
traceroute to 2025:fa9:99:1::2 (2025:fa9:99:1::2), 30 hops max, 80 byte packets
 1  2025:fa9:100:1::1 (2025:fa9:100:1::1)  1.081 ms  1.054 ms  1.038 ms
 2  2025:fa9:102::a (2025:fa9:102::a)  3.225 ms  3.444 ms  3.686 ms
 3  2025:fa9:102::11 (2025:fa9:102::11)  6.959 ms  13.572 ms  16.877 ms
 4  2025:fa9:99:1::2 (2025:fa9:99:1::2)  11.844 ms  19.883 ms  19.878 ms
```

7. IS-IS (Intermediate System to Intermediate System)

- Configured IS-IS for link-state routing in a service provider-style network
- Implemented Level-1 and Level-2 routing for hierarchical design
- Troubleshot routing inconsistencies caused by mismatched IS-IS levels
- Verified route propagation and metric calculations across the network
- Analyzed scalability and efficiency of IS-IS in large networks



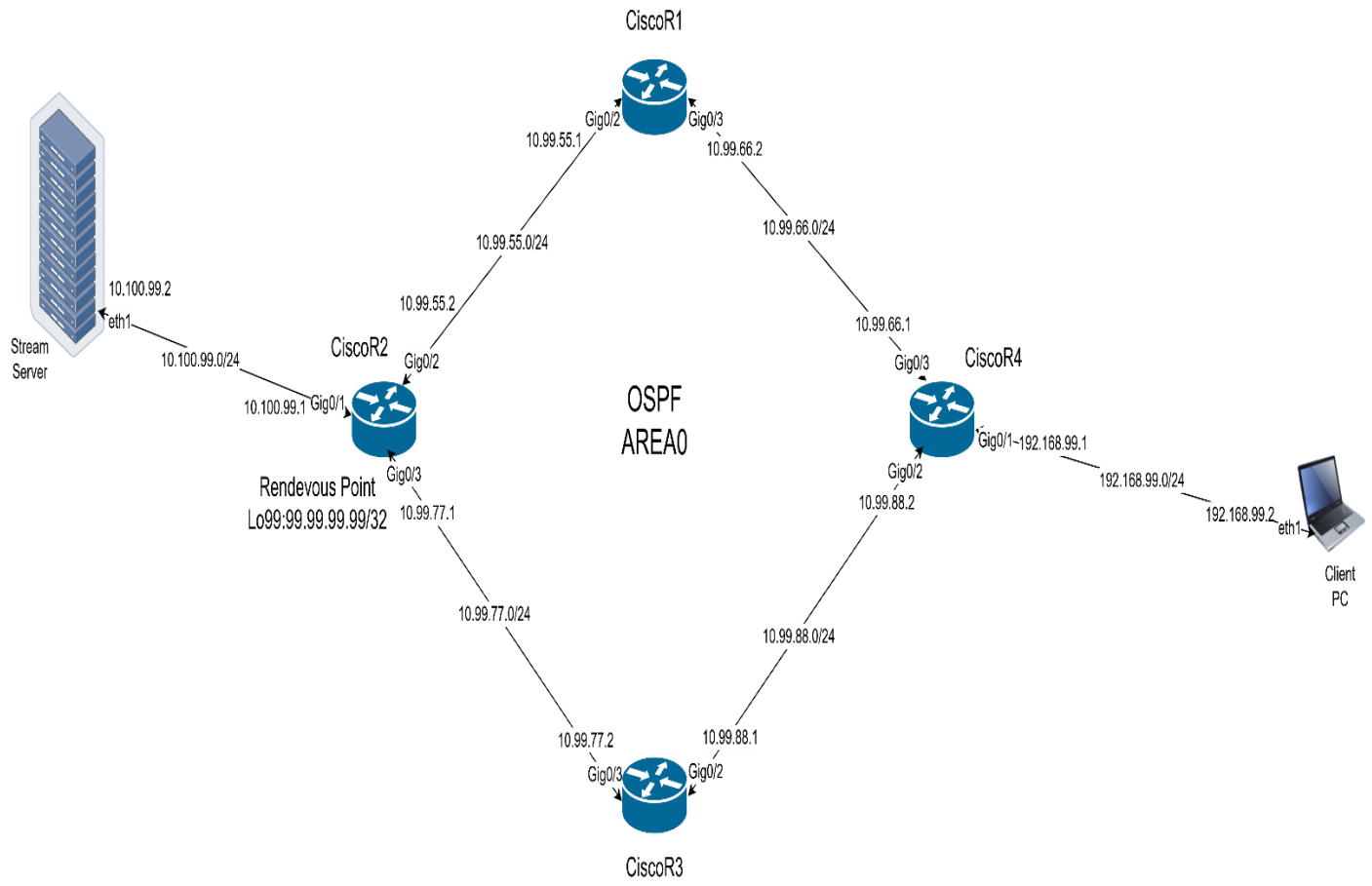
Host A to Host B

```
mint@Host-A:~$ ping 192.168.99.130
PING 192.168.99.130 (192.168.99.130): 56 data bytes
64 bytes from 192.168.99.130: icmp_seq=0 ttl=59 time=5.953 ms
64 bytes from 192.168.99.130: icmp_seq=1 ttl=59 time=6.049 ms
64 bytes from 192.168.99.130: icmp_seq=2 ttl=59 time=6.413 ms
64 bytes from 192.168.99.130: icmp_seq=3 ttl=59 time=6.834 ms
64 bytes from 192.168.99.130: icmp_seq=4 ttl=59 time=6.766 ms
^C--- 192.168.99.130 ping statistics ---
5 packets transmitted, 5 packets received, 0% packet loss
round-trip min/avg/max/stddev = 5.953/6.403/6.834/0.359 ms
```

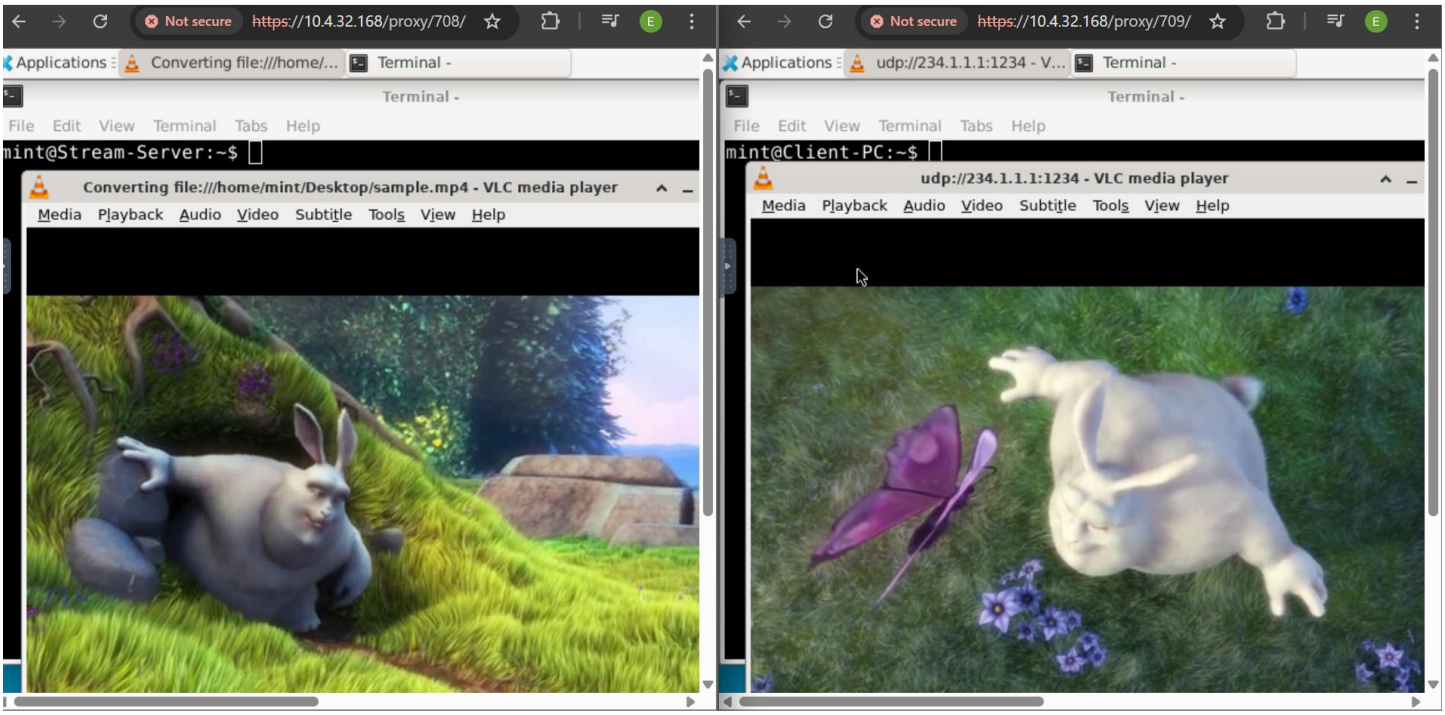
```
mint@Host-A:~$ ping6 -n 2025:fa9:b:99::2
PING 2025:fa9:b:99::2 (2025:fa9:b:99::2): 56 data bytes
64 bytes from 2025:fa9:b:99::2: icmp_seq=0 ttl=59 time=14.627 ms
64 bytes from 2025:fa9:b:99::2: icmp_seq=1 ttl=59 time=7.320 ms
64 bytes from 2025:fa9:b:99::2: icmp_seq=2 ttl=59 time=7.020 ms
64 bytes from 2025:fa9:b:99::2: icmp_seq=3 ttl=59 time=6.779 ms
^C--- 2025:fa9:b:99::2 ping statistics ---
4 packets transmitted, 4 packets received, 0% packet loss
round-trip min/avg/max/stddev = 6.779/8.937/14.627/3.291 ms
```

8. Multicast Routing

- Configured multicast routing using PIM to enable efficient one-to-many communication
- Verified multicast group membership and traffic flow across the network
- Troubleshot multicast failures caused by missing or incorrect PIM configuration
- Validated multicast functionality using VLC-based video streaming
- Analyzed dependency between multicast and underlying unicast routing



Capture the VLC video streaming windows of both PCs, side by side



Lab Environment and Tools

The laboratory exercises documented in this portfolio were conducted using containerlab, a modern network emulation platform that enables the deployment of realistic, container-based network topologies. This environment was selected to closely replicate enterprise and service provider networks while supporting multivendor interoperability within a single, scalable lab setup. The lab topology included routing platforms running Cisco IOS, Juniper Junos, Nokia SR OS, Nokia SR Linux, and Linux-based routing stacks, all instantiated as containers and interconnected using containerlab. Each device was configured using its native command-line interface, reinforcing vendor-specific syntax while maintaining a strong focus on vendor-neutral networking concepts.

Linux-based host containers were used to generate and validate network traffic. Network behavior was verified using ICMP reachability tests, routing table inspection, and protocol-specific validation commands. These tests ensured correct operation of Layer 2 and Layer 3 forwarding across the topology. Multicast functionality was validated using VLC media streaming between host containers to demonstrate efficient one-to-many traffic delivery. This allowed verification of multicast routing behavior at both the control plane and data plane levels.

The containerlab environment supported rapid iteration, troubleshooting, and validation, enabling configuration errors and design issues to be identified and corrected efficiently. The use of containerlab reflects modern network engineering practices and demonstrates familiarity with tools commonly used in network automation, testing, and large-scale lab environments.